

Prompt Calibration Detail

adapted visual feature H + class prompt bank $T \rightarrow$ calibrated emotion logits

Inputs

Adapted visual feature

$$H \in \mathbb{R}^{B \times D}$$



Class prompt feature bank

$$T \in \mathbb{R}^{C \times P \times D}$$

Anxiety



Peace



Weariness



Happiness



Anger



C emotion classes

P prompts per class

Image-Text Similarity

H compared with every prompt feature

H

...



...



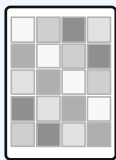
...



T

Prompt-level similarity scores

$$S \in \mathbb{R}^{B \times C \times P}$$



Learnable Prompt Weights

softmax over prompts within each class

w_1



...

w_2



...

w_3



...

w_4



...

$\vdots$

w_t



...

Learnable prompt logits (normalized by softmax)

$$\omega \in \mathbb{R}^{C \times P}$$

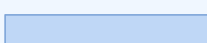
Class Score Aggregation

weighted sum over prompt scores

Anxiety



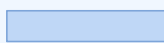
Peace



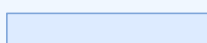
Weariness



Happiness



Anger



class-level scores

$$s \in \mathbb{R}^{B \times C}$$

Scale, Bias, Logits

calibrate each emotion class

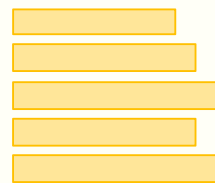
Learnable class scale γ

+

Learnable class bias b

$$\text{logits} = \gamma \cdot \text{score} + b$$

Calibrated emotion logits



C classes

similarity(h, T) $\rightarrow$ softmax prompt weights $\rightarrow$ class scores $\rightarrow$ class scale + bias $\rightarrow$ emotion logits